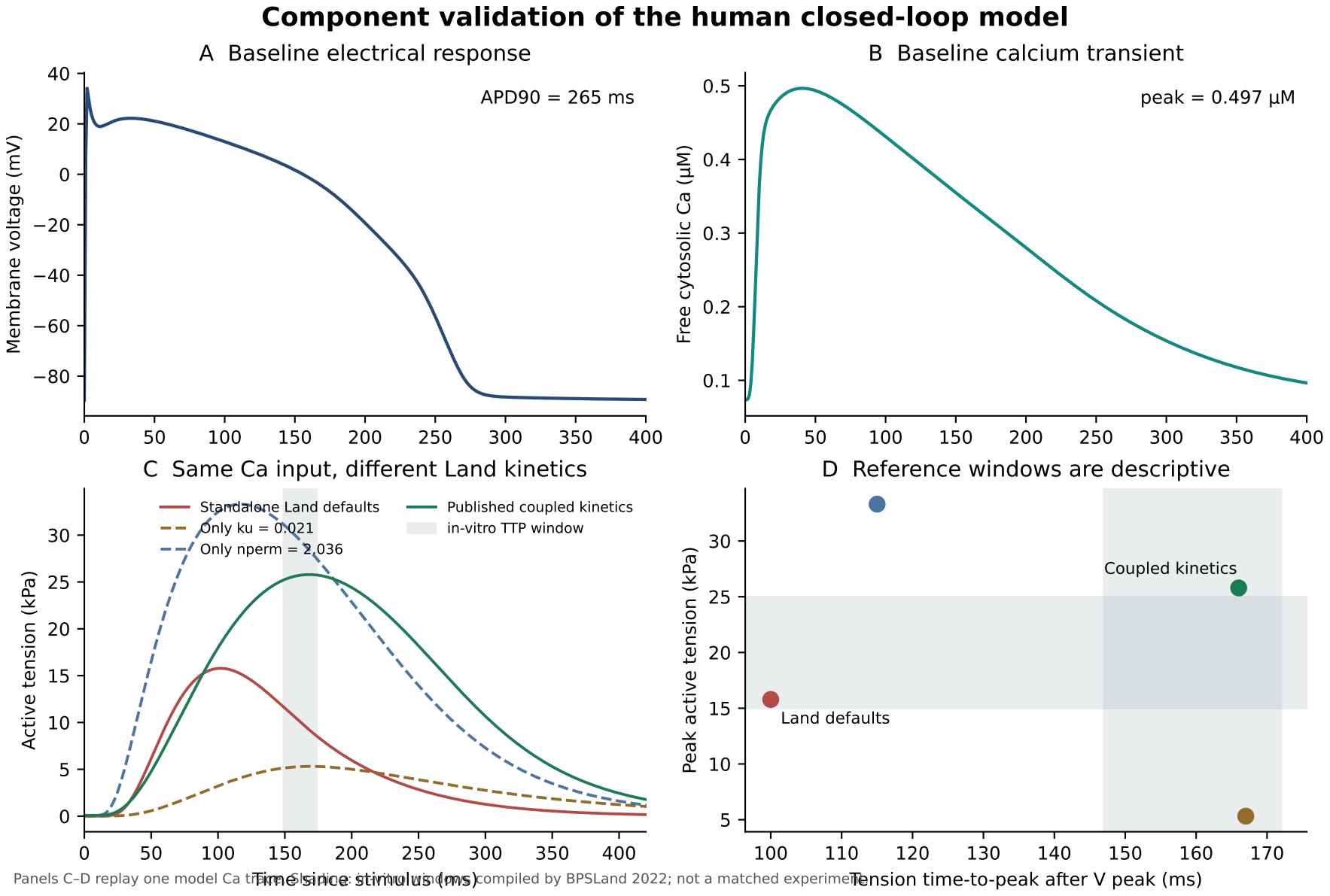
Component validation of the human closed-loop model
A Baseline electrical response
APD90 = 265 ms
Membrane voltage (mV)
B Baseline calcium transient
peak = 0.497 µM
Free cytosolic Ca (µM)
C Same Ca input, different Land kinetics
Standalone Land defaults
Only ku = 0.021
Only nperm = 2.036
Published coupled kinetics
in-vitro TTP window
Active tension (kPa)
Time since stimulus (ms)
D Reference windows are descriptive
Coupled kinetics
Land defaults
Peak active tension (kPa)
Tension time-to-peak after V peak (ms)
Panels C–D replay one model Ca transient: illustrative, compiled by BPSLand 2022; not a matched experiment.